



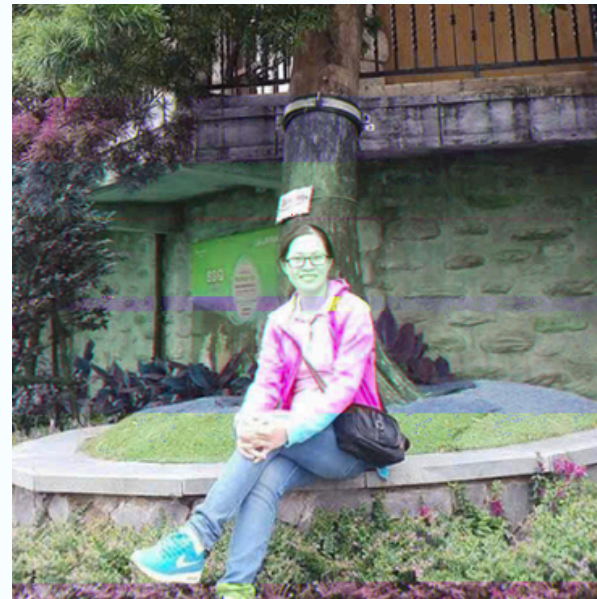
JCAP-DEVELOPMENT OF AN AI-POWERED WEB PLATFORM FOR SCENARIO-BASED JAPANESE CONVERSATION PRACTICE WITH ADAPTIVE LEARNING

By Group SE_54

Da Nang, August 2026



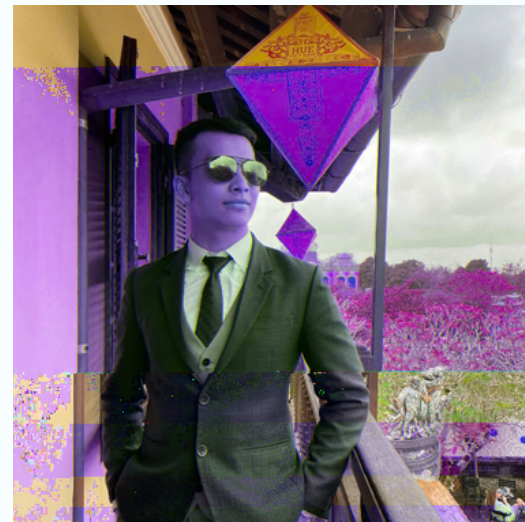
GROUP MEMBERS



Mentor
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Leader
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Member
Nguyễn Quang
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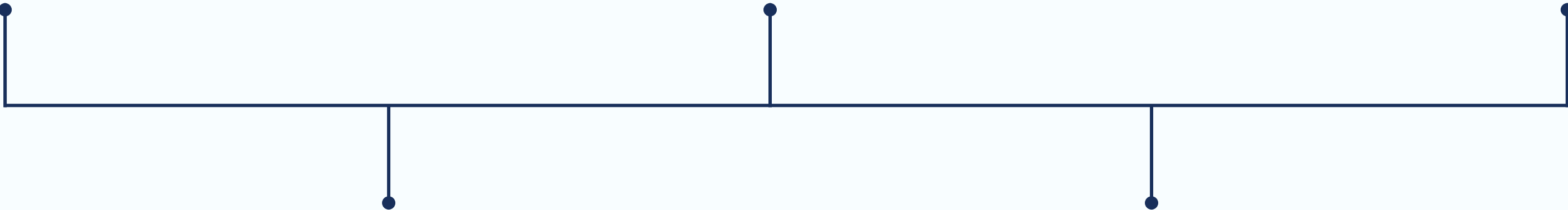


Member
Nguyễn Xuân Trường



Member
Nguyễn Đình Mạnh

TABLE OF CONTENTS



A diagram showing a horizontal line with five vertical lines extending upwards from it, each ending in a dot. These dots are positioned below the text of items 01, 02, 03, 04, and 05, indicating a sequential flow or connection between the items.

01	Project Introduction	03	Use Cases Design	05	Database Design & ERD
02	Main Features & Project Management Plan	04	Screen Flow		

BACKGROUND

- The growing demand for Japanese learning reflects the need to use Japanese beyond textbooks and classrooms.
- However, knowing Japanese does not always mean being able to communicate effectively in real-life situations.
- Developing conversation skills requires practice with realistic scenarios, appropriate responses, and meaningful feedback.
- JCAP is designed to support this journey through AI-powered, scenario-based Japanese conversation practice.



PAIN POINTS

Limited opportunities for realistic conversation practice

- Learners may know vocabulary and grammar but lack opportunities to apply them in real situations.

Limited access to immediate and structured feedback

- Learners may not know what to improve in pronunciation, vocabulary, grammar, or sentence construction.

Different learners, different needs

- A one-size-fits-all practice activity may not effectively address each learner's weaknesses.



EXISTING SYSTEM

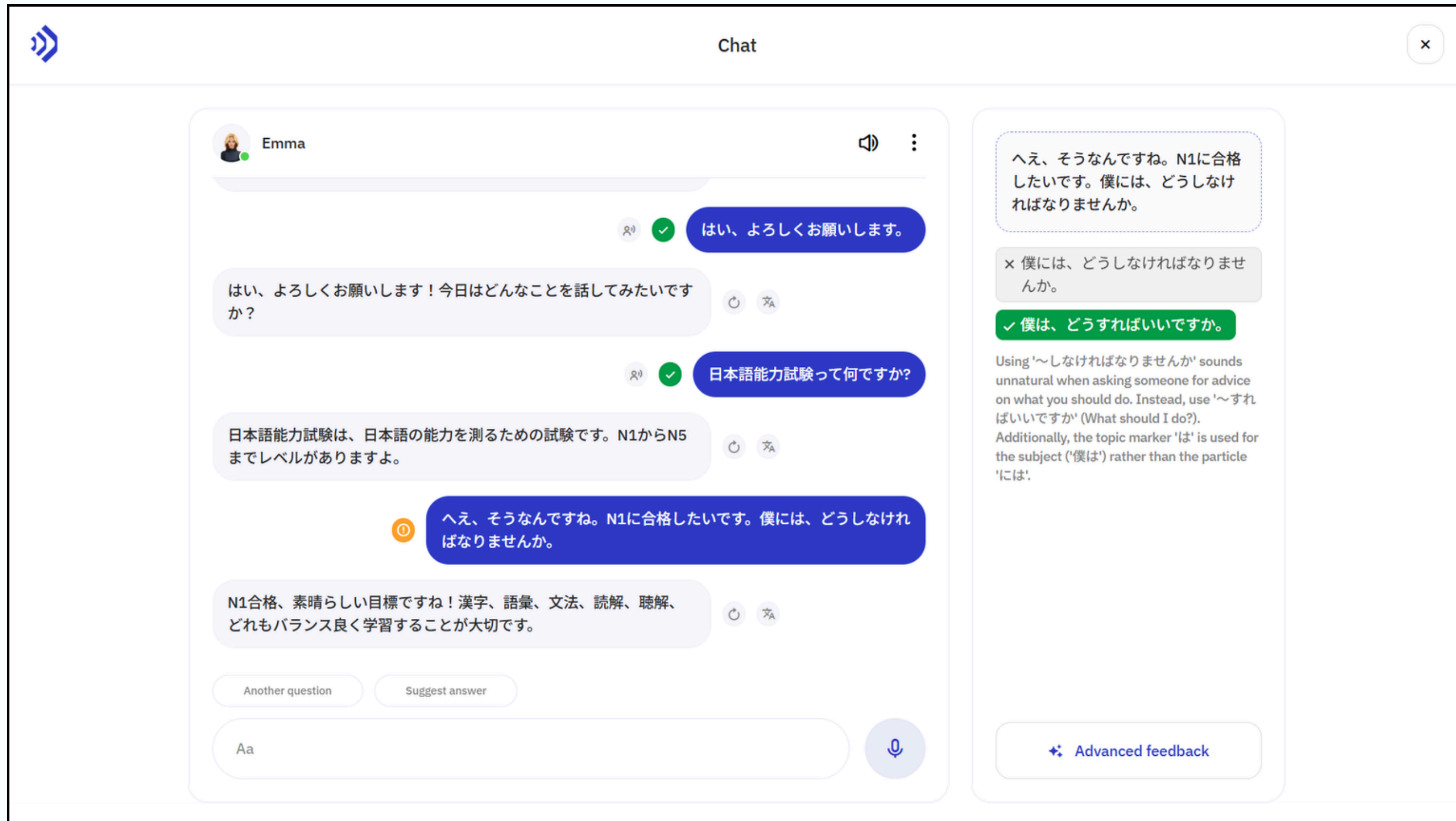
Gliglish



1. Gliglish

- AI-powered free-form conversation with **text and voice input**, supported by **grammar and vocabulary feedback**.
- Users can also receive **suggested replies** and **save useful words or phrases** for later review.

EXISTING SYSTEM



2. Talkpal

- AI-powered conversation through **text and voice**, with **multiple structured learning modes** such as Dialogue, Sentence, Word, and Photo Mode.
- It also supports **roleplay in real-life scenarios** and provides **contextual feedback** on grammar and vocabulary.

THE GAP & OUR SOLUTIONS

Existing systems provide many powerful capabilities, but these capabilities are often experienced as individual activities.

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MAJOR FEATURES

FE-01

FE-02

FE-03

FE-04

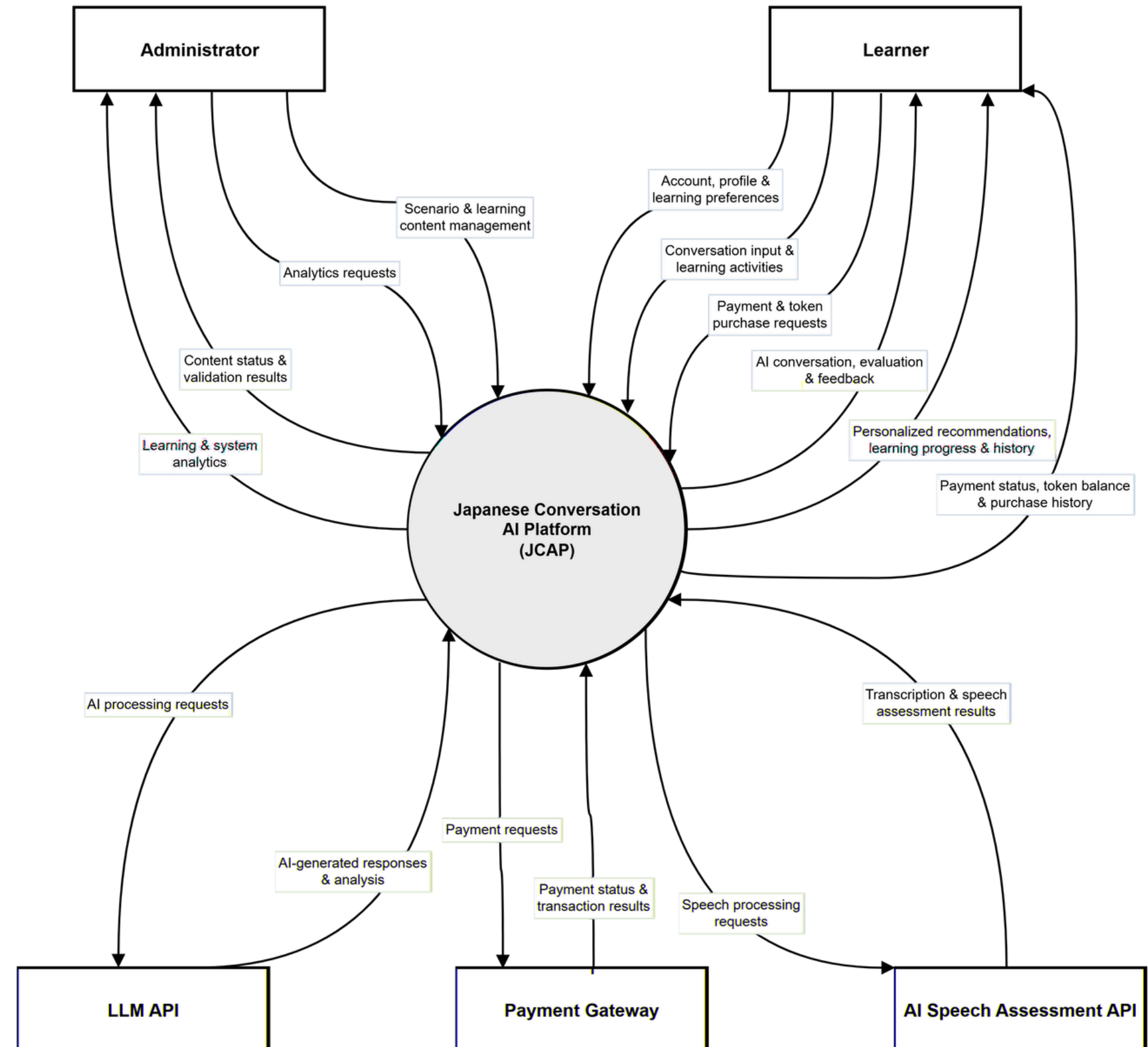
FE-05

FE-06

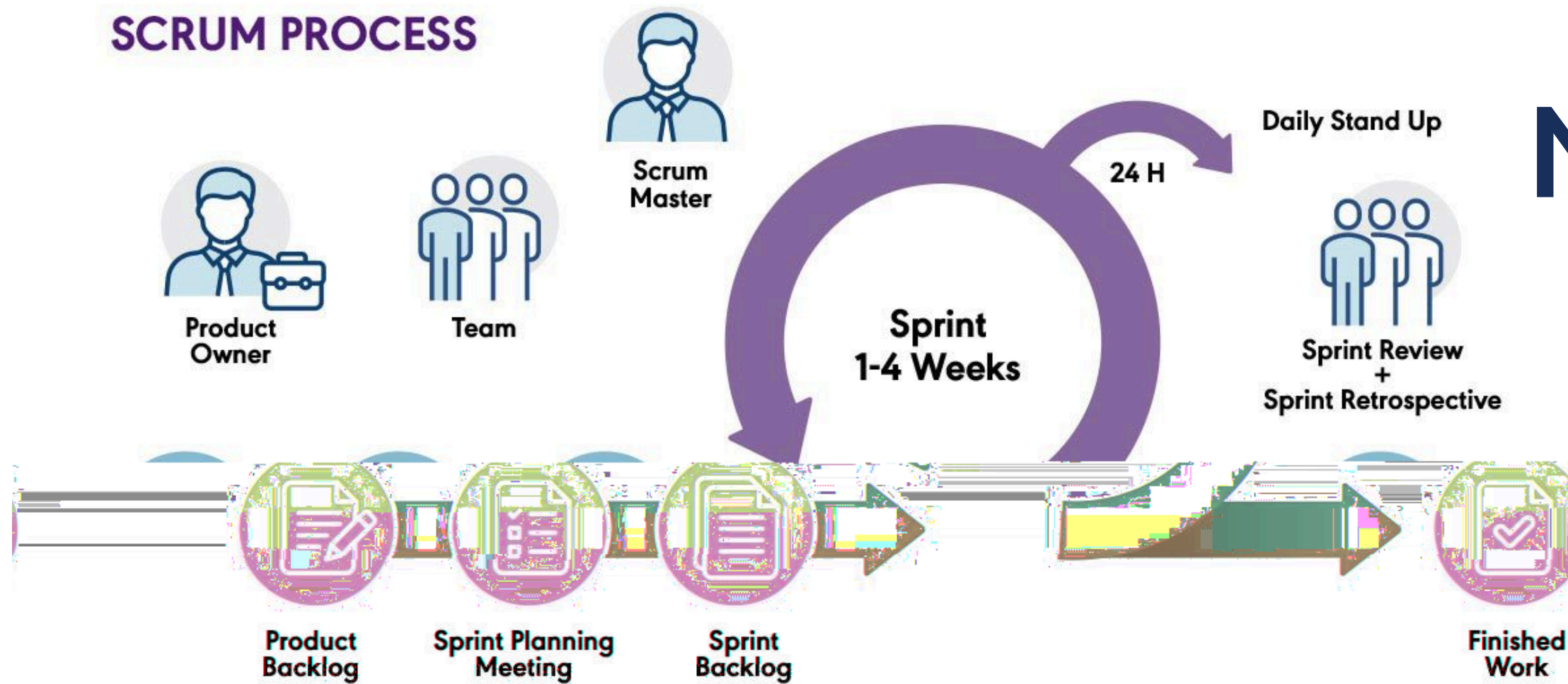


CONTEXT DIAGRAM

System boundary & external interactions



PROJECT MANAGEMENT



18 Weeks
August 01 - December 07,
2026
Sprint 0 — 4 weeks
Sprint 1-7 — 2 weeks each

TASK MANAGEMENT



Jira

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•

•

Spaces

SEP490_FA26_Group_JS ...

Summary List Board Backlog Development Forms Timeline Docs Reports +

Search backlog Filter

Backlog (12 work items) 0 0 0 Create sprint

Create a whiteboard to plan your work Try

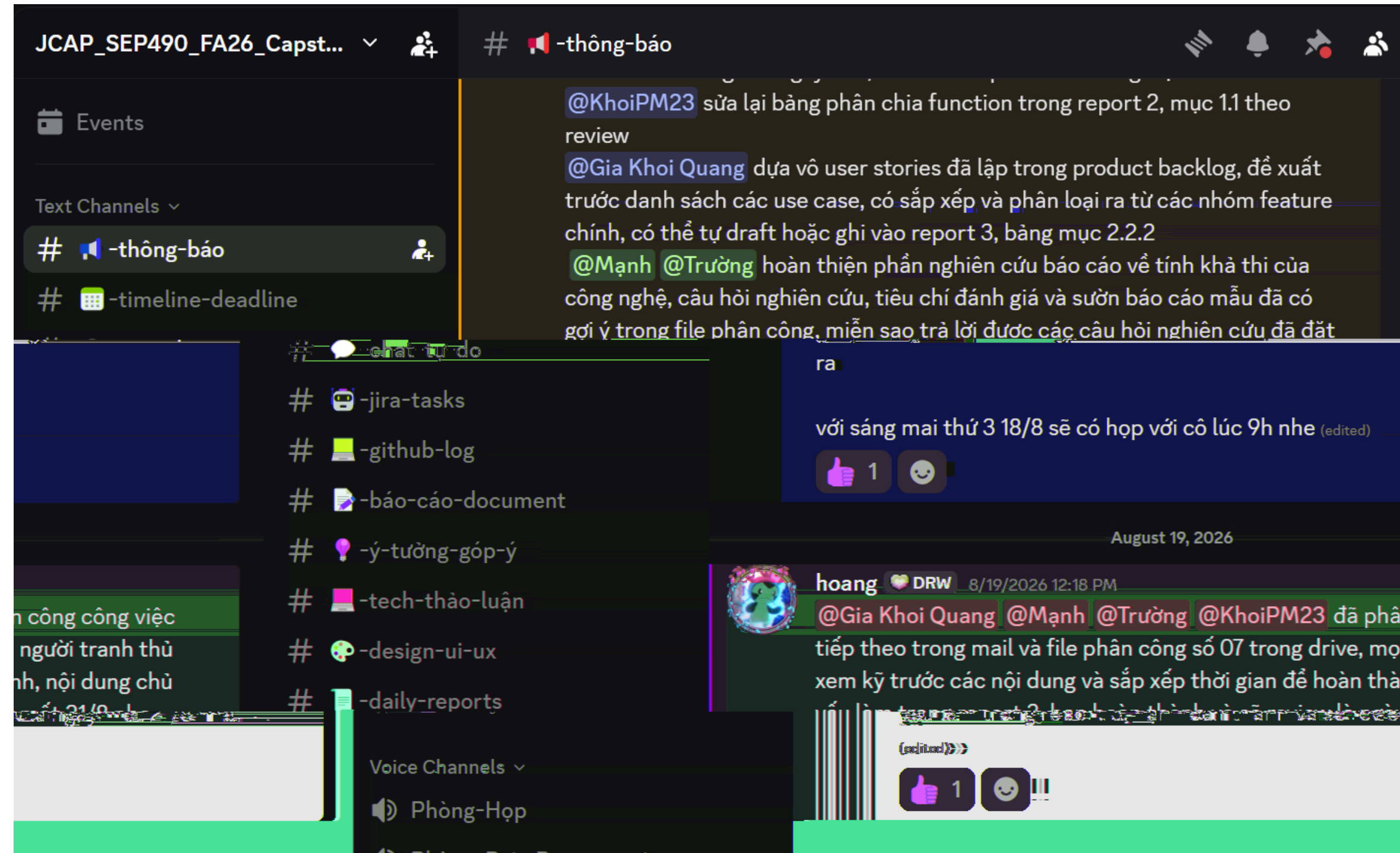
FA26JS-14	Draw Use Case Diagram for section 2.2.1 of Report 3	In Review	Aug 26	-	=	GQ
FA26JS-13	List and briefly describe the main Use Cases for section 2.2.2 of Report 3	In Review	Aug 23	-	=	GQ
FA26JS-12	Specify the list of actors for section 2.1 of Report 3	In Review	Aug 23	-	=	GQ
FA26JS-21	Propose Quality Management approaches from section 2.2 of Report 2	To Do	Aug 23	-	=	L
FA26JS-20	Propose Quality Attributes from section 4.2 of Report 3	To Do	Aug 23	-	=	L
FA26JS-19	Propose Business rules for FE-01, FE-05, FE-06, FE-07 and write them in section 5.1 of R...	To Do	Aug 23	-	=	L
FA26JS-18	Propose Business rules for FE-02, FE-03, FE-04 and write them in section 5.1 of Report 3	To Do	Aug 23	-	=	L
FA26JS-17	Draw Screen Flow for section 3.1.1 of Report 3	To Do	Aug 26	-	=	L
FA26JS-16	Describe Screen authorization in section 3.1.3 of Report 3	To Do	Aug 23	-	=	L
FA26JS-15	Specify the possible screens and categorize into main features, for section 3.1.2 of Repo...	To Do	Aug 23	-	=	L
FA26JS-23	Propose Entities Description and ERD in section 3.1.5 of Report 3	To Do	Aug 26	-	=	L

COMMUNICATION



Discord

Team Communication
& Coordination



DOCUMENT MANAGEMENT



Google Drive

Centralized Project
Documentation

Shared with me > SEP490 SE 54 Document...

Ask Gemini

Name	Type	People	Modified	Source
FSoft Artifact				
Nội dung phân công và...				
SEP490 Report				

FA26_Nguyen Thi Anh...

SEP490...

TRƯỜNG ĐẠI HỌC FPT
CAPSTONE PROJECT REGISTER

Class: CP_SEP490 Duration time: From 01/09/2024 To 31/12/2024
Professor: Information Technology Specialty: Software Engineering
Kind of person make register: Lecturer ☐ Student ☐

1. Register information for supervisor

Full name	Phone	E-Mail	Title
Supervisor 1	0933881437	Doanx143@fpt.edu.vn	CTP
Supervisor 2			

2. Register information for student

Full name	Student ID	Phone	E-Mail	Specialty
Nguyen Thi Anh...	0933881437	0933881437	Doanx143@fpt.edu.vn	Software Engineering

The Student's Guide of Capstone Project Document for the FPT University Students

1. General Introduction

1.1. Capstone Project's description

Develop of a significantly complex software system, employing knowledge gained from courses throughout the program including defining requirements, design, implementation, and quality assurance. Students must follow a suitable project model, pay attention to quality issues, and manage the project themselves, following all appropriate project management techniques. Success of the project has to deliver the stages on time.

Sample deliverables:

Students should deliver one or several iterations of a software system, along with all artifacts appropriate to the process model they are using. These could daily include a project plan (perhaps updated regularly) and the following: On requirements, risk analysis, division of the work into tasks, milestones, requirements (including use cases, architectural and design documents, test plan, source code, and executable software package).

CODE MANAGEMENT



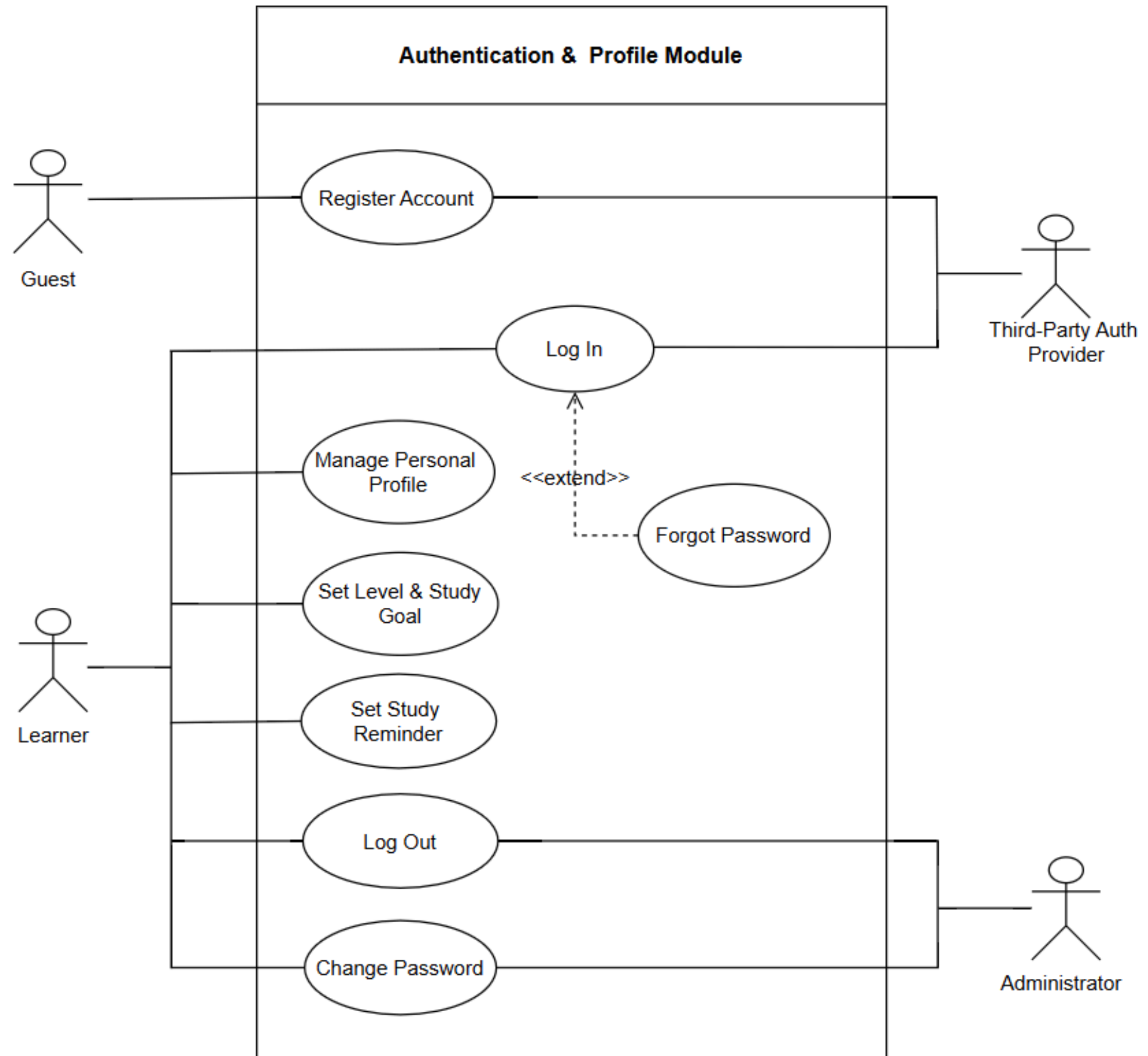
GitHub

Version Control &
Collaborative
Development



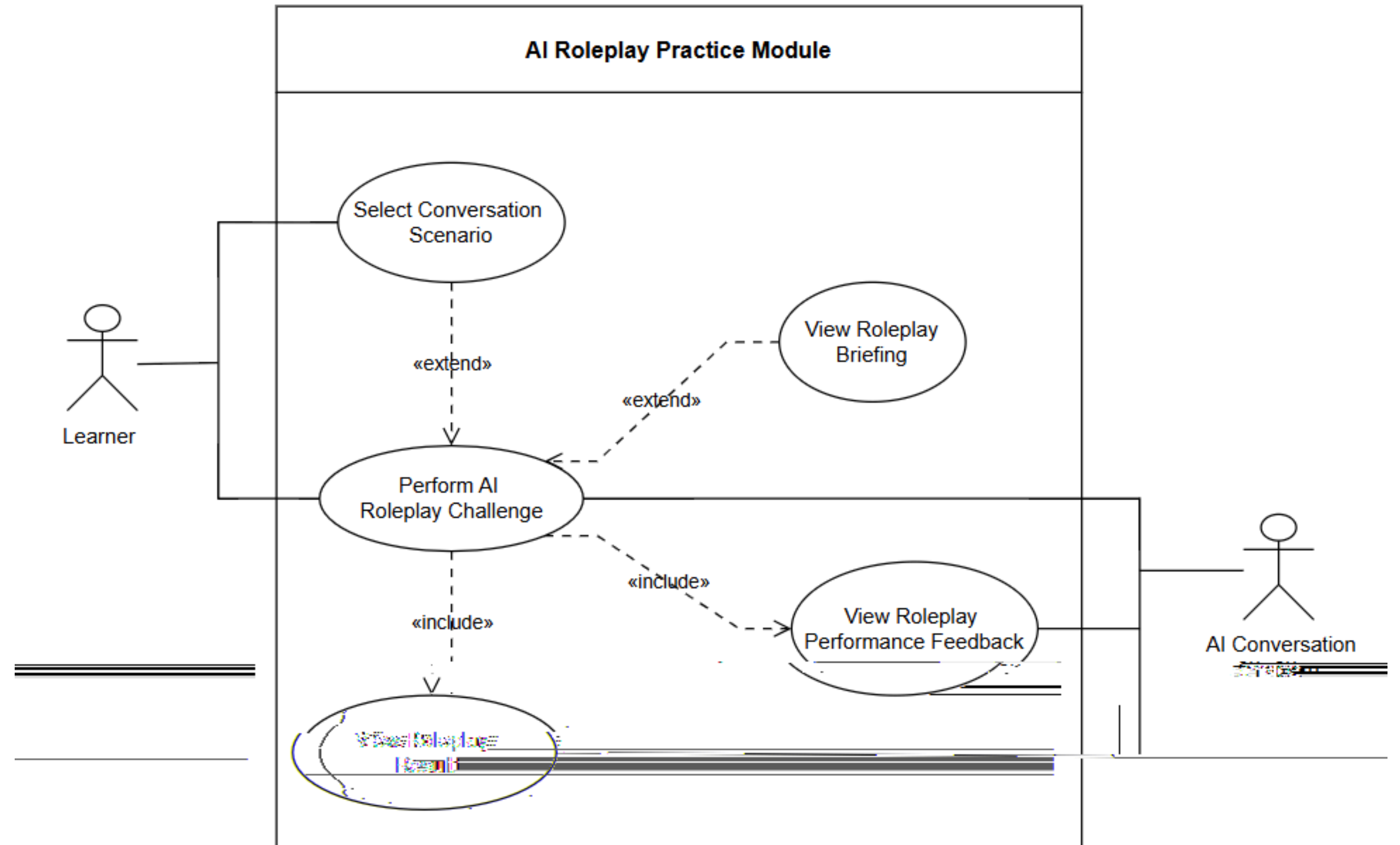
USECASE DIAGRAM

Authentication & Profile Module



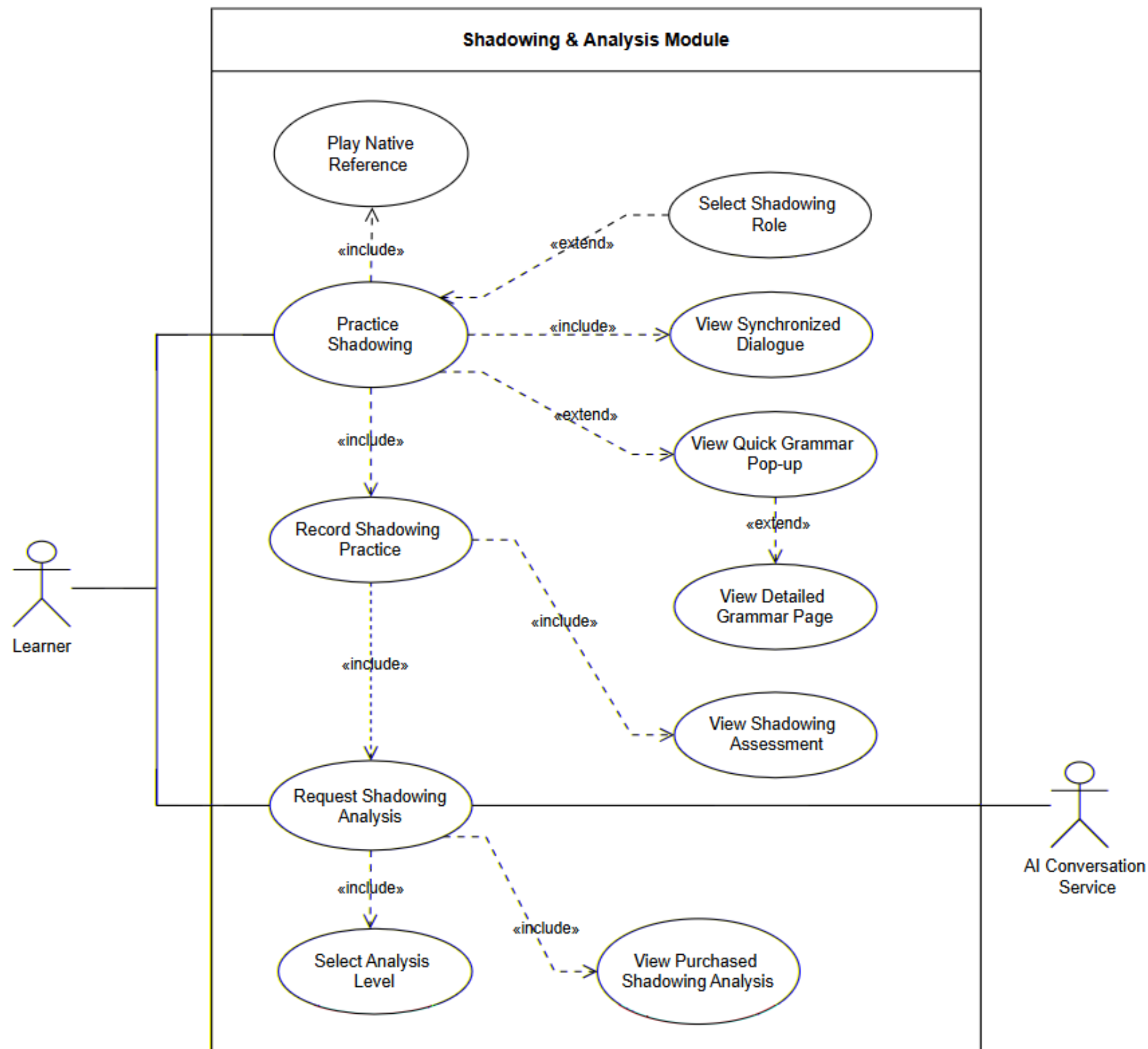
USECASE DIAGRAM

AI Roleplay Practice Module



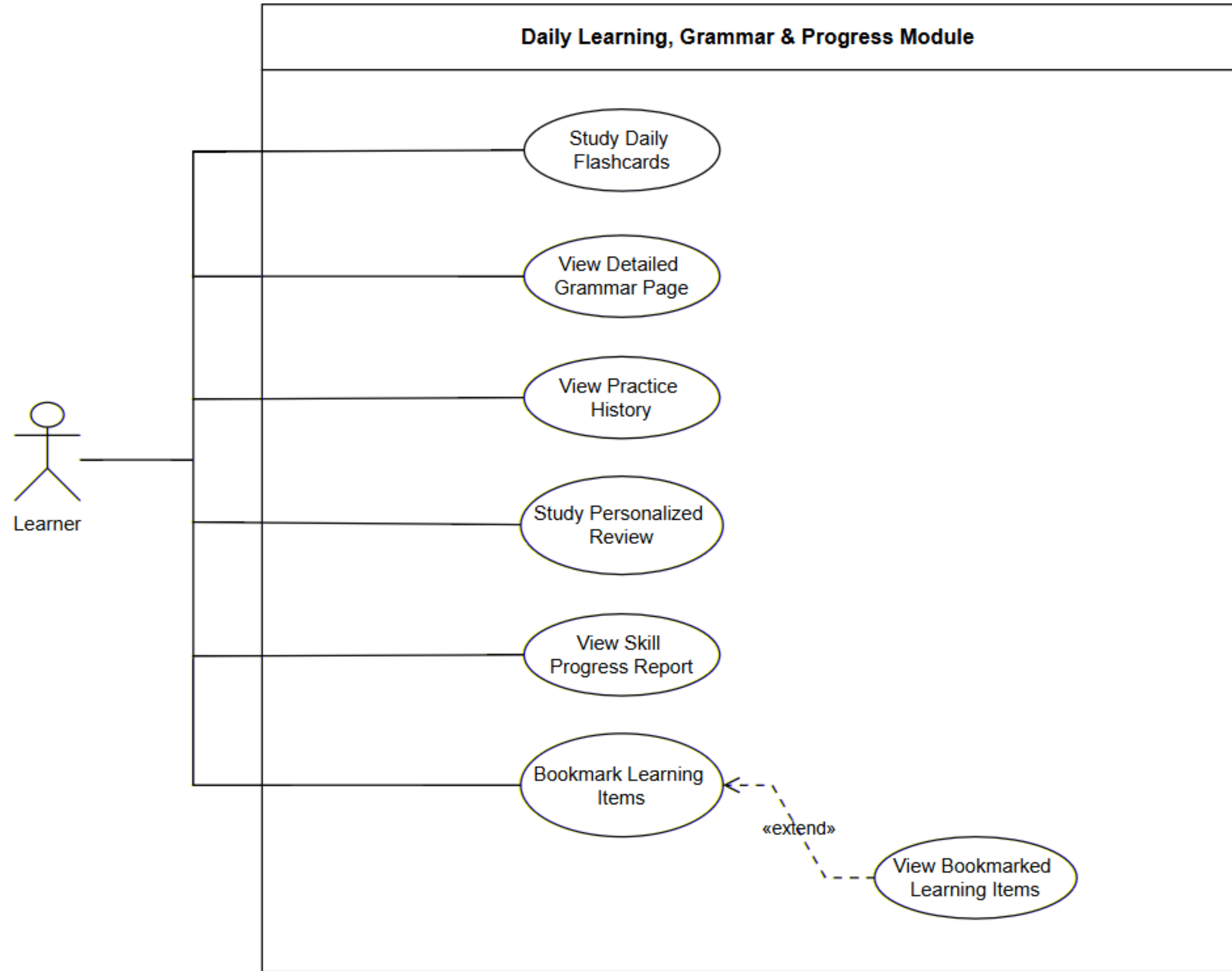
USECASE DIAGRAM

Shadowing & Analysis Module



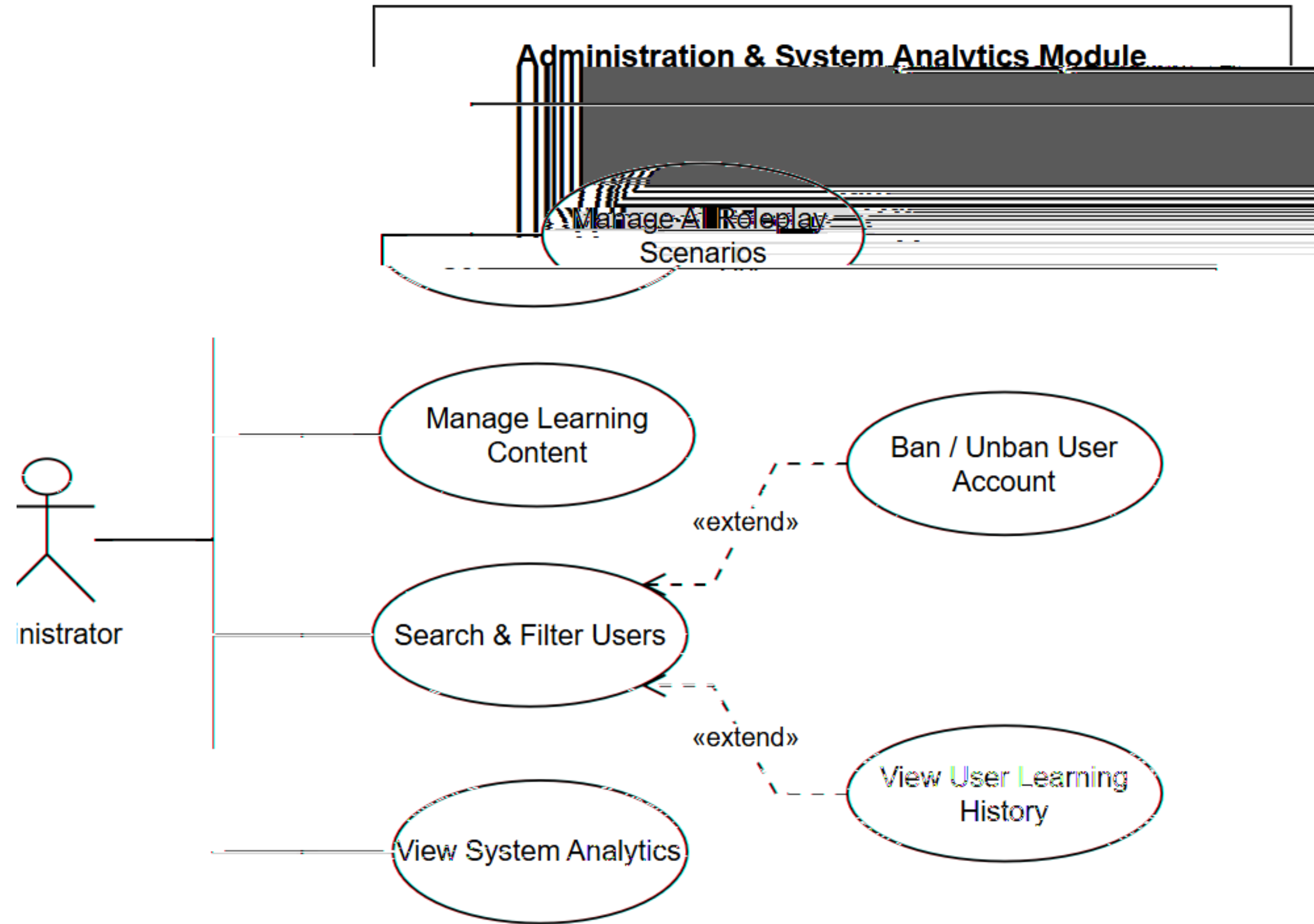
USECASE DIAGRAM

Daily Learning, Grammar & Progress Module



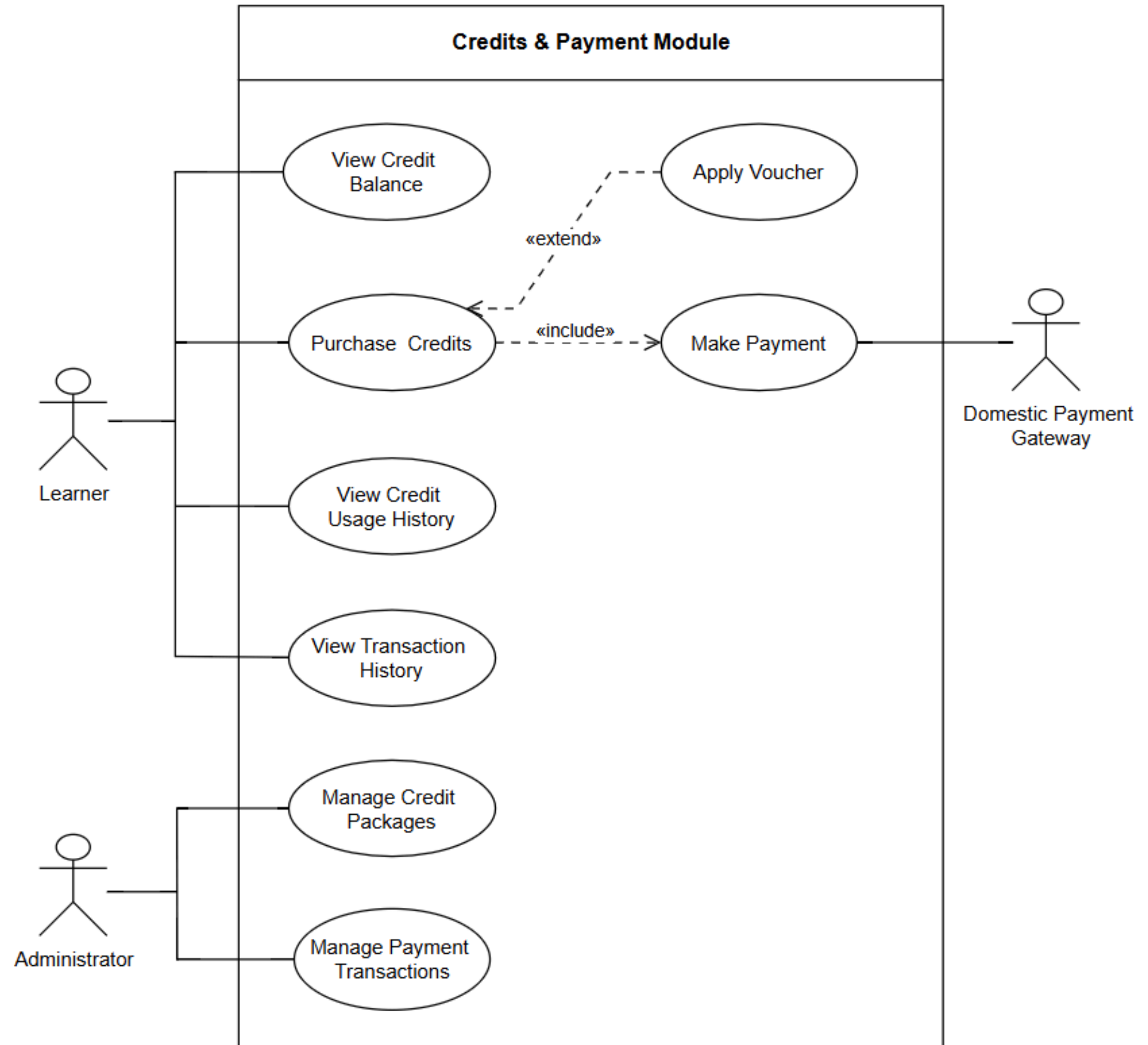
USECASE DIAGRAM

Administration & System Analytics Module



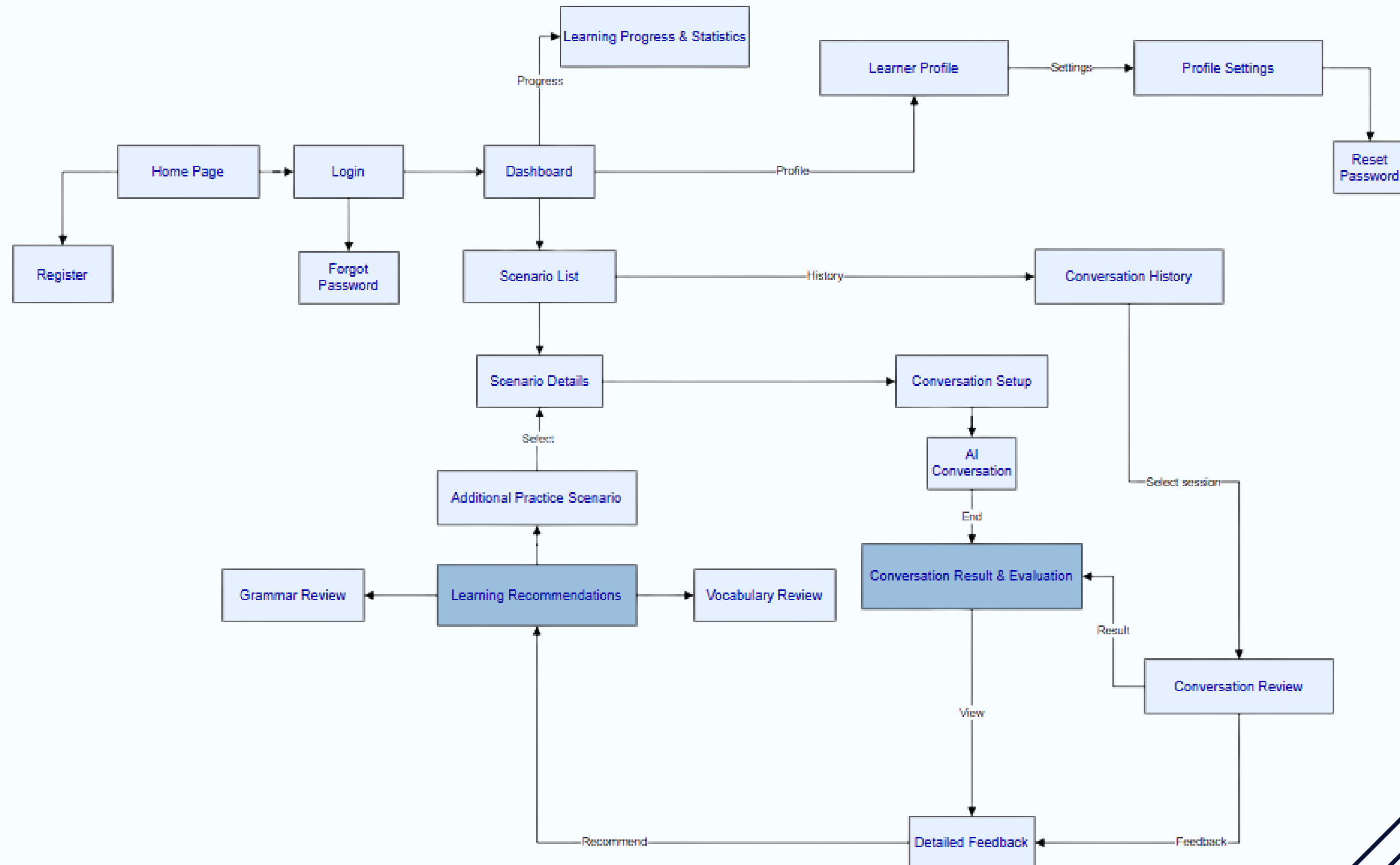
USECASE DIAGRAM

Credits & Payment Module

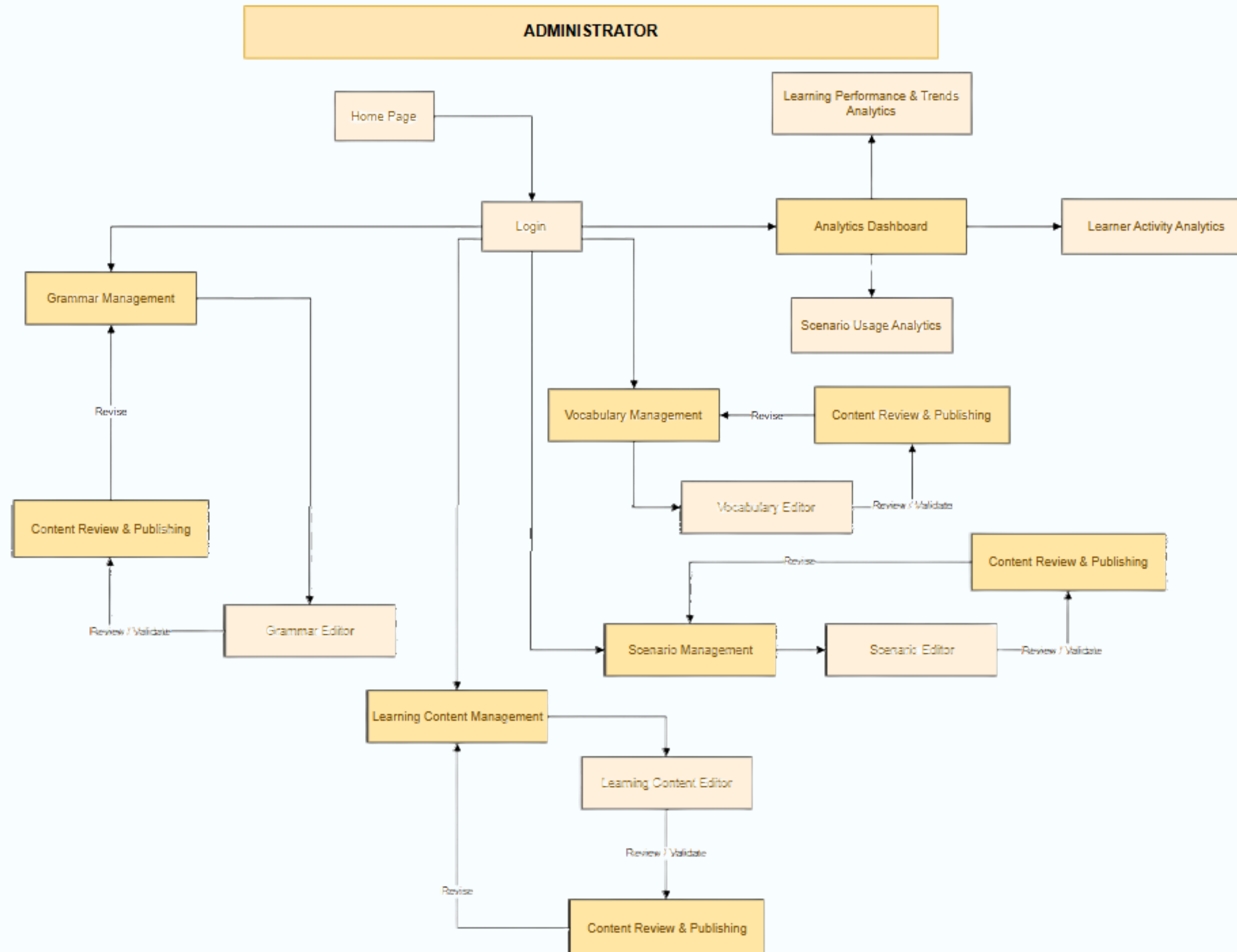


SCREEN FLOW

LEARNER

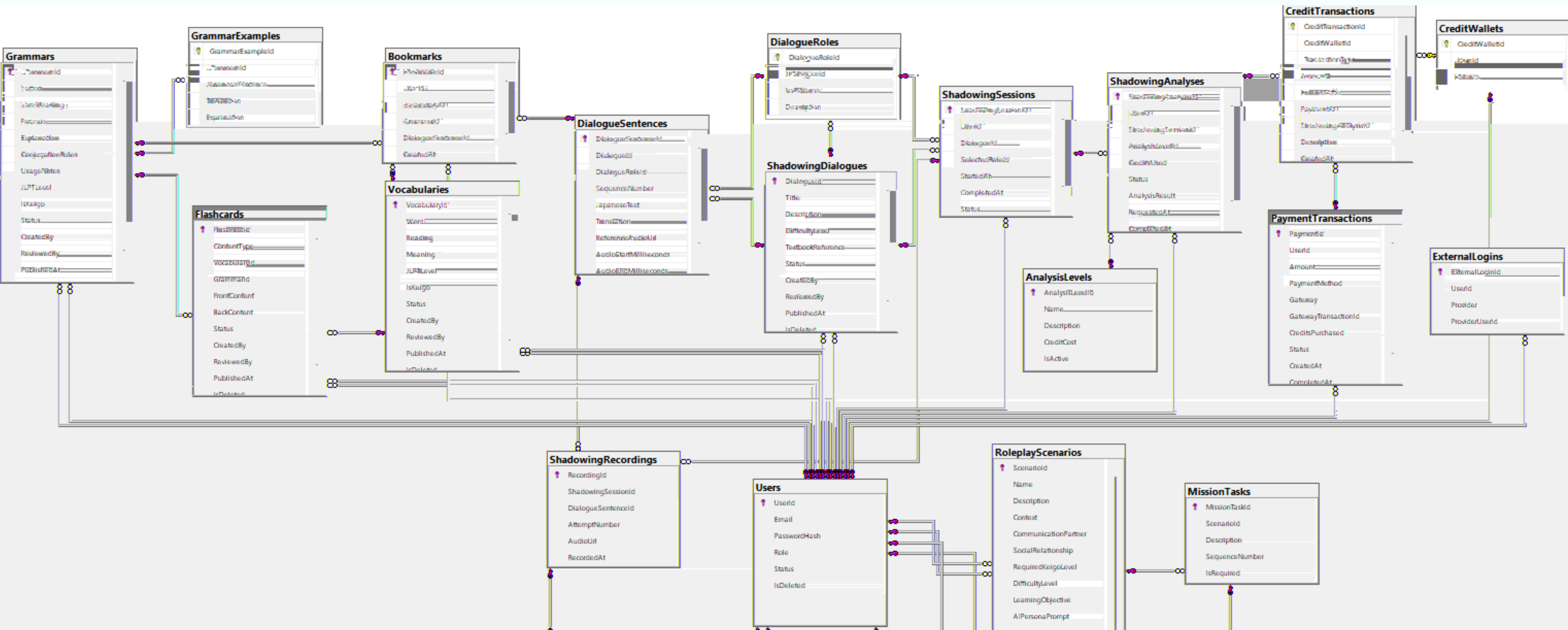


SCREEN FLOW

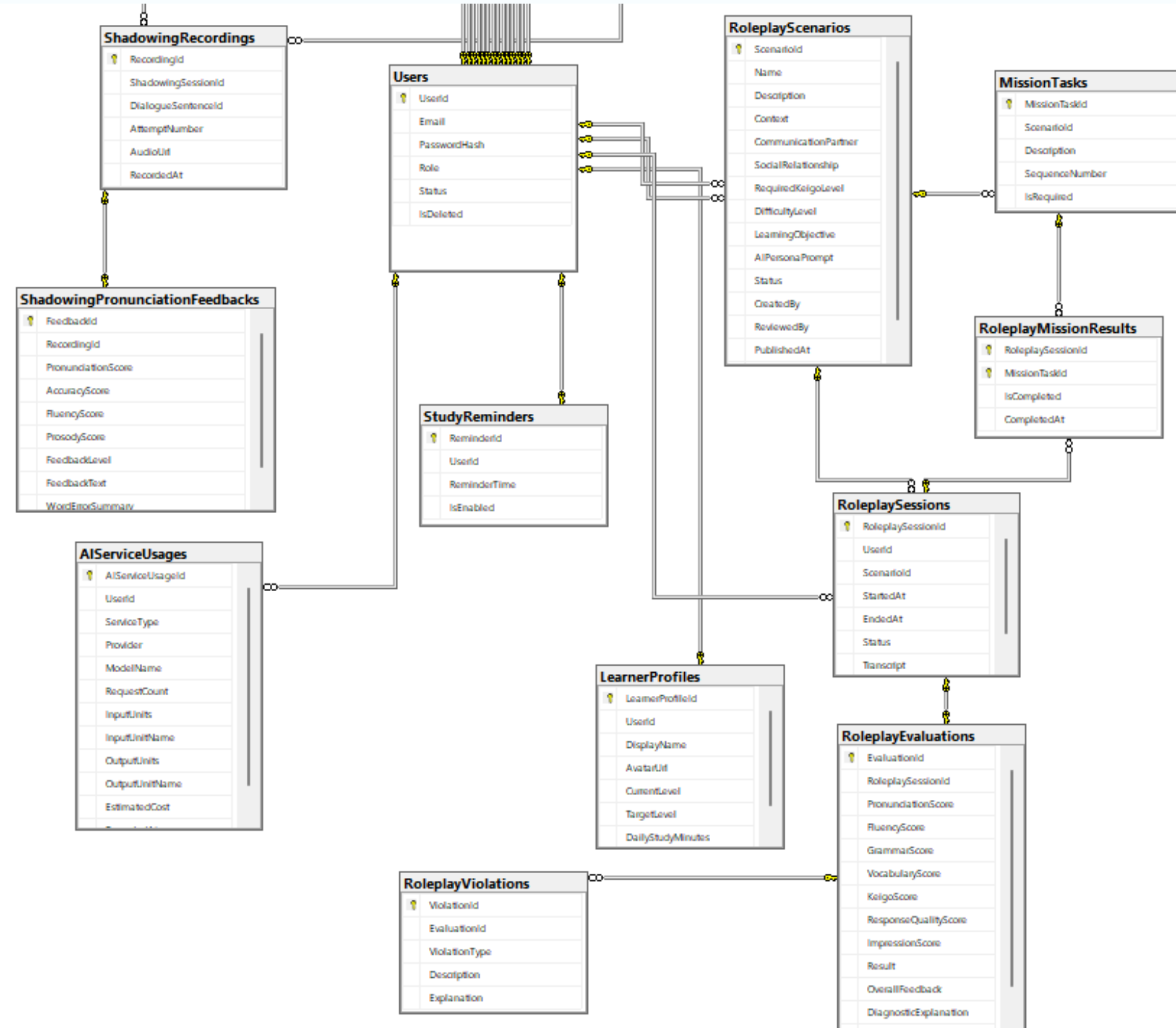




ERD



ERD



The image features a light blue background with four decorative corner elements. Each corner contains a dark blue square with diagonal lines and a medium blue square. The top-left and bottom-right corners have the dark blue square in the foreground, while the top-right and bottom-left corners have the medium blue square in the foreground.

THANK YOU